

Ahmad Mahmud

PHD STUDENT

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Education

University of Strasbourg

PHD IN COMPUTER SCIENCE

Strasbourg, France

Nov 2023 - present

- **Thesis:** Software Defined Low-Power and Lossy Wireless Networks
- **Advisors:** Julien Montavont & Thomas Noel
- **Grant:** MESR grant of the Doctoral School of Mathematics, Information Sciences and Engineering (MSII)

University of Calabria

MS IN TELECOMMUNICATION ENGINEERING (GPA: 110 CUM LAUDE/110)

Calabria, Italy

Oct 2021 - Oct 2023

- **First rank** over the 2 years of the degree.
- **Thesis:** Improving the Quality of Federated Learning using Programmable Networks
- **Advisors:** Pasquale Pace & Antonio Iera
- **Grant:** International Master Scholarship of the University of Calabria

Latakia University (formerly: Tishreen University)

BS IN COMMUNICATIONS AND ELECTRONIC ENGINEERING (GPA: 93/100)

Latakia, Syria

Sep 2014 - Sep 2019

- **First rank** over the 5 years of the degree.
- **Thesis:** Millimeter Waves (mmWs) Channel Simulation using MATALAB
- **Advisor:** Ghadir Madi

Professional Experience

2023-Now **PhD Student**, University of Strasbourg, ICube laboratory, France

2019-2021 **Workshop Trainer in Telecommunications & Programming**, Skills and Career Center, Latakia University (formerly: Tishreen University), Syria

Publications

INTERNATIONAL JOURNALS

Ahmad Mahmud, Julien Montavont, and Thomas Noel. 2026. FINE-STACK: Flexible architecture for low-power lossy IoT Networks with programmable network STACK and semantic runtime updates. *Elsevier Ad Hoc Networks*.

Ahmad Mahmud, Julien Montavont, and Thomas Noel. 2025. PENSIL: Programmable Network Stack for low-power lossy IoT networks using Lightweight-virtualization. *Elsevier Internet of Things*.

Ahmad Mahmud, Pasquale Pace, and Antonio Iera. 2024. The Role of SDN to Improve Client Selection in Federated Learning. *IEEE Communications Magazine*.

INTERNATIONAL CONFERENCES

Ahmad Mahmud, Julien Montavont, and Thoms Noel. 2025. Programmable Solutions for Low-power Lossy Wireless Networks: A Study of SDN and Femto Containers. *International Conference on Advanced Information Networking and Applications (AINA)*. Springer Nature Switzerland.

Ahmad Mahmud, Pasquale Pace, and Antonio Iera. 2024. SDN-assisted client selection to enhance the quality of federated learning processes. *Proceedings of the IEEE Wireless Communications and Networking Conference (WCNC)*.

Ahmad Mahmud, Pasquale Pace, and Antonio Iera.. 2023. Improving the quality of federated learning processes via software defined networking. *NetAISys '23: Proceedings of the 1st International Workshop on Networked AI Systems*.

NATIONAL CONFERENCES

Ahmad Mahmud, Julien Montavont, and Thomas Noel. 2026. Divide and conquer: a semantic and modular reconfiguration of the network stack for constrained wireless networks. *CoRes 2026, Accepted*.

Ahmad Mahmud, Julien Montavont, and Thomas Noel. 2024. Menu or a la carte? An architecture for programming the data plane of constrained wireless networks. *CoRes 2024*.

ARTICLES UNDER REVIEW

Ahmad Mahmud, Julien Montavont, and Thomas Noel. 2026. AI-Enabled Traffic-Aware Closed-Loop Framework for MAC-Layer Control in Multi-Hop IoT Networks. *MSWiM 2026 Conference*.

Teaching Experience

Wireless Networks

UNIVERSITY OF STRASBOURG - M1 (BAC+4)

Strasbourg, France

Spring 2025

- Volume: 20h (Lectures (CM) + Exercises (TD) + Practical sessions (TP))
- Course Responsible: Ahmad Mahmud (me) & Fabrice Theoleyre

Internet of Things

UNIVERSITY OF STRASBOURG - M2 (BAC+5)

Strasbourg, France

Fall 2024 & Fall 2025

- Volume: 36h (Practical sessions (TP))
- Course Responsible: Julien Montavont

Supervision

2026 **Charbel Gergès**, Master-1 Research Project (TER), *Reliable and Efficient Over-the-Air Software Update Mechanisms for Wireless Networks* (Bac+4).

University of
Strasbourg

Awards & Grants

2023 **Excellence Prize for First Graduate**, University of Calabria
2022 **Excellence Prize for First-ranked Student**, University of Calabria
2021 **International Master Scholarship**, University of Calabria

2019 **Excellence Certificate for First Graduate**, Latakia University
2018 **Excellence Certificate for Academic Distinguish (Rank: First)**, Latakia University
2017 **Excellence Certificate for Academic Distinguish (Rank: First)**, Latakia University
2016 **Excellence Certificate for Academic Distinguish (Rank: First)**, Latakia University
2015 **Excellence Certificate for Academic Distinguish (Rank: First)**, Latakia University

Technical Skills

PROGRAMMING	C/C++, Python, Java, MATLAB, Linux Bash
MACHINE LEARNING	scikit-learn, PyTorch, TensorFlow
SDN	RYU Controller, OpenDayLight Controller, POX Controller
EMBEDDED SYSTEMS	RIOT, Contiki-NG, Zephyr
TOOLS	FIT IoT-LAB, Cooja, Mininet/Mininet-WiFi, NS3

Projects

Flexible Network Architecture for LLWN

UNIVERSITY OF STRASBOURG

2023-2026

- Used Tools: C/C++, Python, RIOT OS
- Code: [Repository 1](#) & [Repository 2](#)

Closed-loop ML-based reconfiguration of LLWN

UNIVERSITY OF STRASBOURG

2023-2026

- Used Tools: Machine-Learning, Data Analysis, C/C++, Python, RIOT OS
- Code: Soon (waiting for publication acceptance)

Extension of RPL protocol in RIOT OS to support Non-storing mode

UNIVERSITY OF STRASBOURG

2025

- Used Tools: C/C++, Git
- Code: [Repository](#)

Improving the Quality of Federated Learning using Programmable Networks

UNIVERSITY OF CALABRIA

2022-2023

- Used Tools: Python, Federated Learning, GNS3, OpenDayLight Controller, OpenFlow

Performance Evaluation of Different Shortest-Path Algorithms in SDN using Different Cost Parameters

UNIVERSITY OF CALABRIA

2023

- Used Tools: RYU Controller, OpenFlow, Networkx Library, Mininet, Dijkstra, Bellman-Ford, A*, Floyd-Warshall.
- Code: [Repository](#)

Load Balancing Technique for Face Recognition Video Streaming in Wireless Networks

UNIVERSITY OF CALABRIA

2023

- Used Tools: RYU Controller, OpenFlow, Networkx Library, Mininet, MininetWifi.
- Code: [Repository](#)

RSNA Screening Mammography Breast Cancer Detection

UNIVERSITY OF CALABRIA

2023

- Used Tools: Machine learning, Deep learning, Computer vision, Image processing, Python, TensorFlow, Kaggle
- Code: [Repository](#)

Detecting contradiction and entailment in multilingual text

UNIVERSITY OF CALABRIA

2022

- Used Tools: Machine learning, Natural language processing, Deep learning, Python, TensorFlow, Hugging Face, Kaggle
- Code: [Repository](#)

Skin Deases Detector App

UNIVERSITY OF CALABRIA

2022

- Used Tools: Android Studio, Java, Python, TFLite, Deep Leaning
- Code: [Repository](#)

Languages

ENGLISH	C1: Read, Written, Spoken
FRENCH	B2: Read, Written, Spoken
ITALIAN	B1: Spoken
ARABIC	Mother Tongue